

Transit Access Inequity in Los Angeles County

By: Brandon Acosta, Fong Lieu, Marcela Lozano, Yarden Sasson, Jessica Thai

Executive Summary

This project investigates transportation equity in Los Angeles County by determining the spatial relationship between socioeconomic variables and proximity to bus stops. The variables taken from the census tract and analyzed included poverty levels, median household income, the poverty probability index, and households without access to a vehicle. These variables were paired with a point dataset of bus stops to better identify where community needs and transportation infrastructure align, and where they do not.

The results of this analysis show a clear density of socioeconomic disadvantage mainly in South Los Angeles, East Los Angeles, and portions of the San Fernando Valley. These neighborhoods display lower median household incomes, high poverty percentages, and a high percentage of households without a car, indicating that they have a higher dependency on public transportation. The density of bus stops is very high in areas like Downtown Los Angeles, but the density decreases as you move away from that area, creating potential inequity in access for the people in areas that need it most. Based on the data, it is recommended that expansion targets specific areas of transit service in order to improve access to transportation and provide transportation equity in underserved communities.

Introduction and Problem Definitions

When we look at the infrastructure of a city, one of the most important factors that people look at is equal access to transportation. Transportation allows the people within a city to be

connected with each other and contributes to economic growth and other opportunities. With the vast differences in economic statuses in Los Angeles County and the high costs associated with owning a vehicle, ensuring easy access to transportation in underserved communities becomes especially important. These high costs can include relatively higher gas prices compared to the rest of the country, high prices for a vehicle resulting from tariffs to bring in the vehicles from overseas, and limited availability of new products, which increases the price due to increased demand. Without equal access to transportation, especially in underserved communities, it limits the opportunities for people to find higher-paying jobs, access to healthcare, poverty, and many other things that people consistently take for granted.

It is important to look at accessibility specifically in transportation because it provides individuals with access to every element of their lives. Without access to transportation, it makes it extremely difficult to go to the grocery store, visit specialized doctors that may not be available in certain areas, go to their job if it is further away, and more. Transportation allows different parts of the city to be connected and provides different opportunities that they may not have locally. It is worth studying because everyone deserves access to opportunities and resources, no matter where they are. People live in certain areas because of factors including income, family size, access to good education, and more. Not everyone can afford to live in an area with top-tier resources, which results in people settling in areas with fewer local services. Equitable transportation helps bridge this gap by giving individuals the ability to reach opportunities beyond their immediate neighborhoods. This also allows for increased economic growth from the business being conducted with people commuting from farther locations.

This report will look at whether areas of various socioeconomic statuses within Los Angeles County have access to safe, affordable, and reliable transportation (a.k.a. Transportation

Equity). This project utilizes spatial relationships to identify specific relationships between variables, including income, poverty, vehicle availability, and transit infrastructure, to determine whether there are communities in higher need of transportation equity. The visualizations conducted in this report will answer the following questions:

1. Where is the highest concentration of socioeconomically vulnerable people within Los Angeles County?
2. Do these communities have enough access to bus stops, or is there a deficit of bus stops based on the amount needed for optimal transit infrastructure?
3. What patterns emerge across income, poverty, and vehicle access that can inform the city planning necessary for public transportation?

With this information, we will be able to uncover insights that could help guide future discussions when it comes to transportation and resource allocation in order to produce equal access to resources that these communities may need.

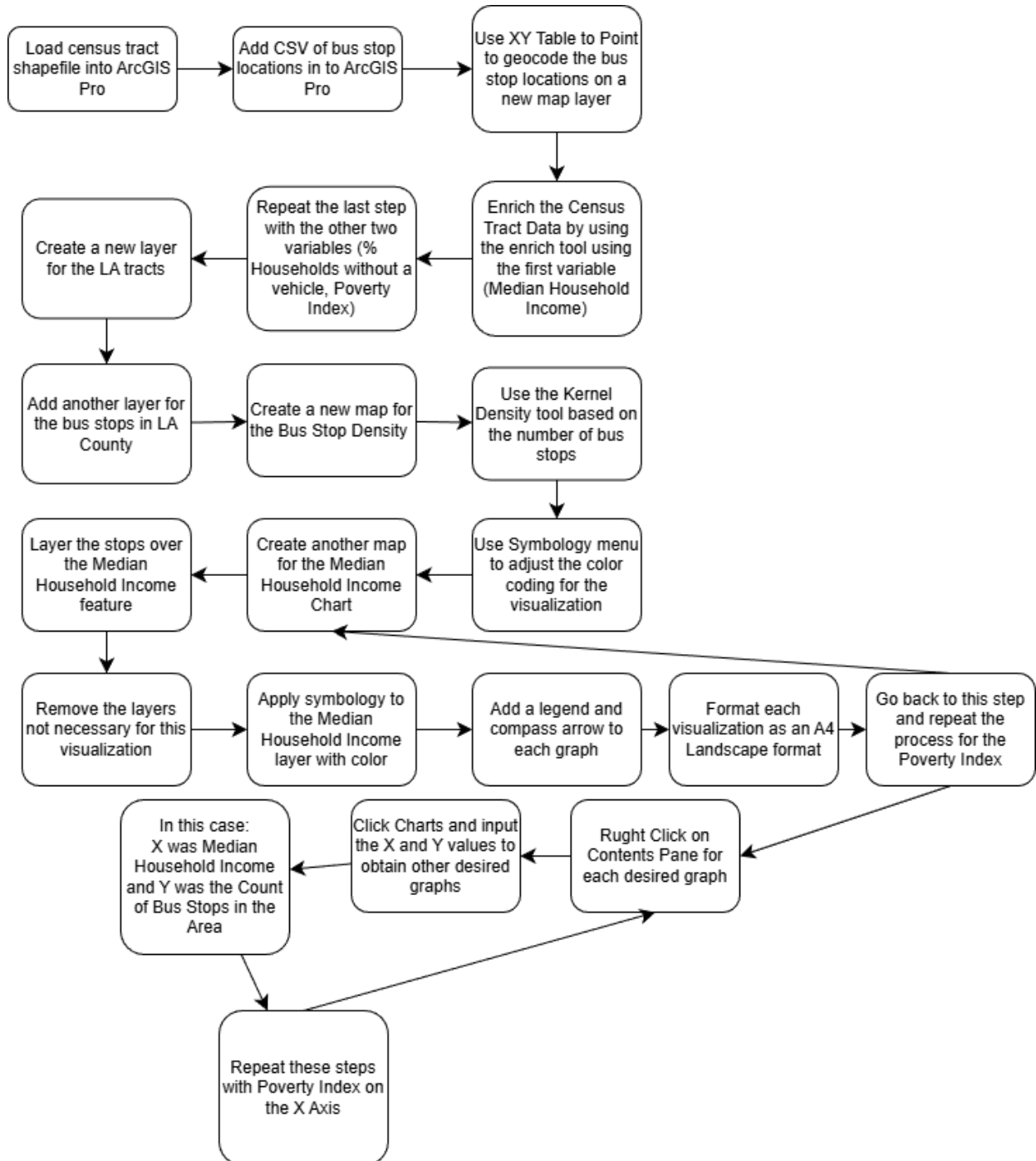
Data Processing

Before beginning the analysis, we collected and organized all of the datasets needed for this project. This section describes the layers we used, where the data came from, and how each dataset was prepared. Laying this out helps show how the information was processed and why it is suitable for the maps and results that follow.

Feature Classes (Layers) Used for Analysis

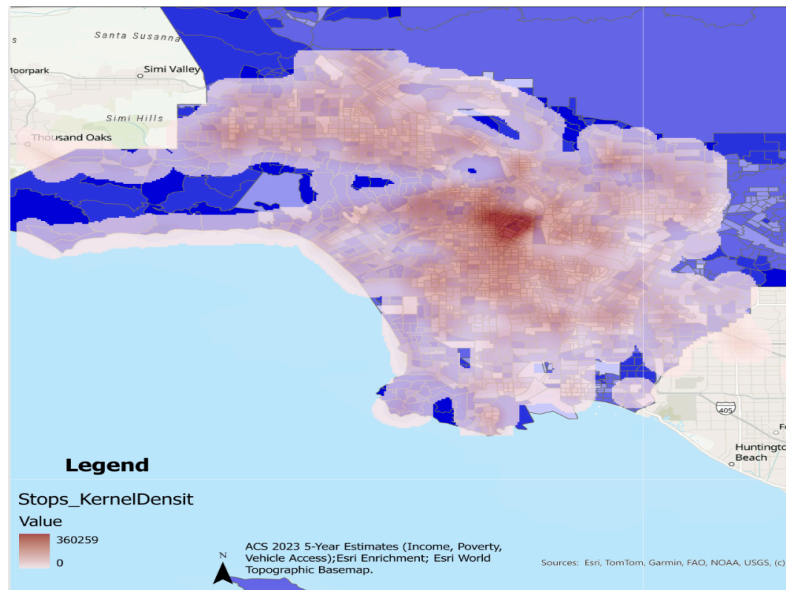
Layer Name	Short Description	Data Source Location	Features (Rows)	Attributes (Columns)	Shape of the Data Structure
LA Metro Bus Stops	Point locations of all LA Metro bus stops	LA Metro GTFS Feed (stops.txt)	11,839	12	Points
Bus Stop Kernel Density Surface	Raster surface showing the concentration of bus stops	Generated in ArcGIS Pro (Kernel Density tool)	58	17	Raster
Median Household Income (ACS 2023)	Median household income by census tract	ACS 2023 via Esri Enrichment	2,495	34	Polygons
Poverty Index (ACS 2023)	Poverty indicators by census tract	ACS 2023 via Esri Enrichment	2,495	34	Polygons
Households Without a Vehicle (ACS 2023)	Households without access to a car by tract	ACS 2023 via Esri Enrichment	2,495	34	Polygons
Los Angeles County Boundary	County boundary used for clipping and analysis	TIGER/Line 2023 (cb_2023_us_county_20m.zip)	85,186	16	Polygons

Visualization Process Flow Chart



Analysis and Results

(1) Bus Stop Density



Looking at this specific map, we can see that the area of Los Angeles County with the highest density is in Downtown Los Angeles. Downtown Los Angeles is where Los Angeles began and where its initial infrastructure was developed. As time moved on and Los Angeles grew, the transportation infrastructure grew alongside it to other areas where many people are living, including the San Fernando Valley, the San Gabriel Valley, the West Valley, and more. The areas of high density include areas with higher density of transportation (similar to Union Station), where it may be harder to get around because of smaller streets or increased traffic, or the population in that area is increasingly dependent on transportation. The lower density areas could also indicate wealthier areas that aren't as reliant on public transportation, suburban areas where it is mainly composed of residential areas, or nature areas (i.e., coastlines, nature preserves, mountain ranges, etc). While this can show us where there are more bus stops, it is hard to determine, solely based on this data, whether these low-density areas actually need more access to transportation or not. In order to get a more accurate picture, there are other factors that need to be considered,

including income, the current state of access that people have to their own transportation, and the current state of the city they are living in.

(2) Households with no Vehicles

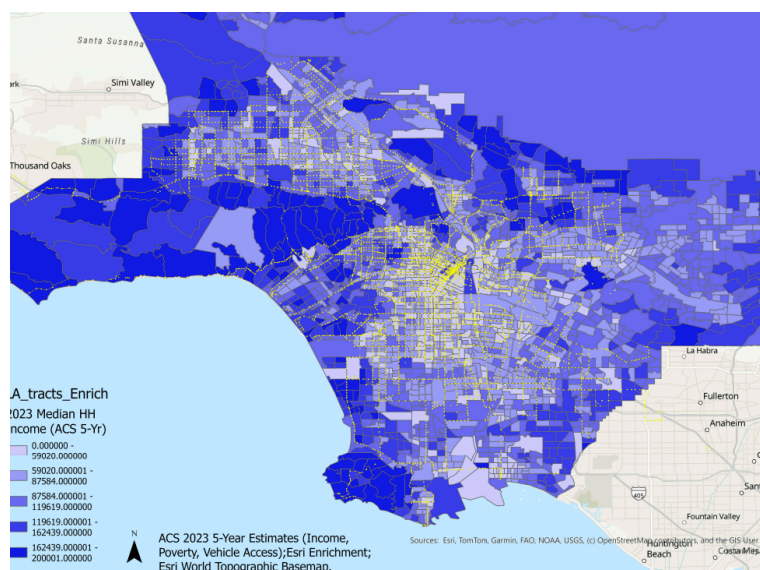


From this map, it is evident that there are areas with clear disparities in vehicle ownership, such as Compton and East Los Angeles. These cities are characterized by a high concentration of residents who do not own or have limited access to personal vehicles, increasing their reliance on public transportation for daily needs. Within Compton, there are several census tracts with more than 80 households without a vehicle, and some even exceed 100. This likely means that many of these residents heavily depend on the bus to commute to work, to attend school, and to access vital services. This is also very similar to the East Los Angeles area, where the density of no-vehicle households is high in many of the census tracts. These commuters are located in some of the most transit-dependent sections of Los Angeles County.

However, when examining the bus stops on the map, the density of the bus stops is significantly lower than that of areas like Koreatown and Downtown Los Angeles. With fewer

bus stops in Compton and East Los Angeles it raises concerns about whether current transit options are meeting community needs. For households without cars, even small limitations in service, such as longer walking distances to stops or infrequent bus arrivals, can create serious barriers. This can lead to late arrivals at work, fewer opportunities to take on multiple jobs, limited access to healthcare appointments, and reduced engagement in community activities. In both Compton and East LA, high rates of zero-vehicle households combined with modest bus stop density highlight an ongoing need for improved transportation equity.

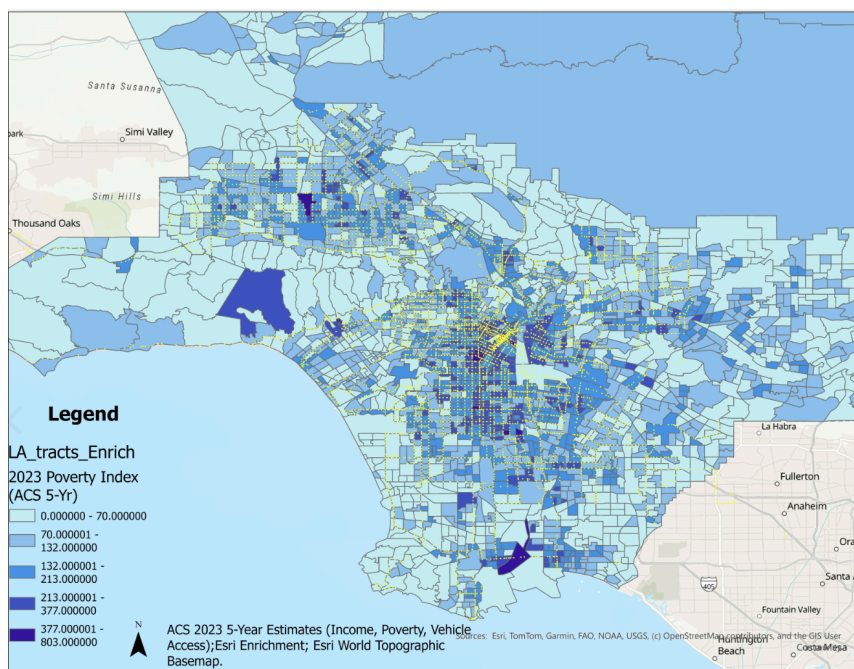
(3) Median Household Income



The median household income map shows that Compton and East Los Angeles are among the lowest-income regions in Los Angeles County. Low median incomes indicate that a lot of the residents have limited financial resources. This can create barriers to owning a car, as there are additional fees that come with it, such as fuel, insurance, maintenance, etc. As a result, public transportation becomes essential in many of these neighborhoods because it gives them the ability to get to work or school without the financial stress. This data also suggests broader

implications for mobility and opportunity. Limited household income can also restrict access to flexible job opportunities, reliable childcare, or healthcare options that require traveling beyond the immediate area. If public transit is not frequent or convenient enough, people may have to face longer travel times that could interfere with their schedules and responsibilities. This creates a cycle in which low income and limited mobility reinforce each other. A more accessible transit system could help break this cycle by connecting the residents in these areas to a wider range of jobs and opportunities. Improving the transit service in low-income areas can help reduce transportation barriers and support economic stability over time.

(4) Poverty



This map shows the poverty index level (more specifically, the Poverty Probability Index) across each part of Los Angeles. This score uses 10 household questions to calculate the percentage of likelihood that a household is poor. This looks at elements including the number of kids going to school, the material your roof is made of, what kind of vehicle they own,

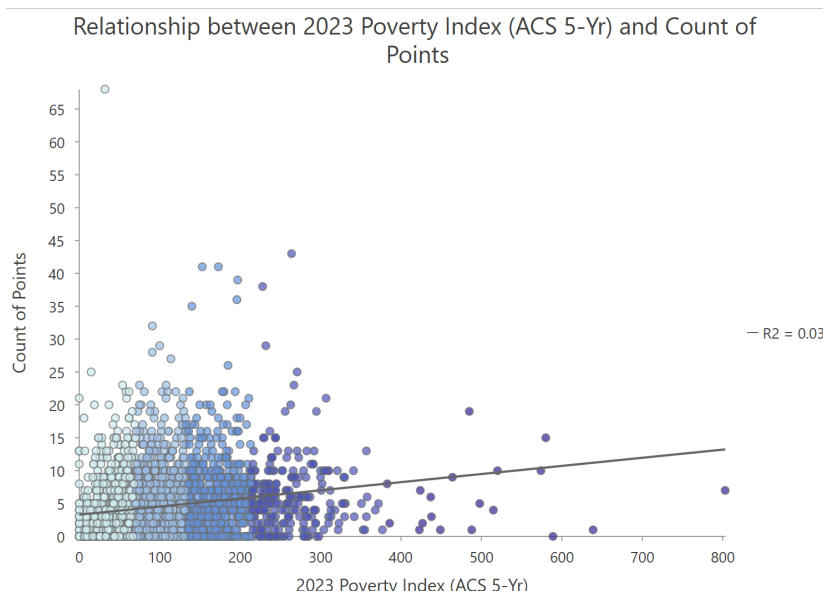
appliances in their household, and more. The answers to this test are then used to match a percentage of poverty that is used for strategic targeting and impact measurement. Based on the map, poverty in Los Angeles County is mostly concentrated in the dense urban areas, more specifically, Central and South LA. These areas include Downtown Los Angeles, Pico-Union, and Central Avenue. These neighborhoods usually have older housing that has not been updated in years or a high amount of multi-family housing, which is more affordable. These areas also have more high-paying jobs, which leads to longer commutes in this area. This is why it is crucial to make sure that there are more bus stations in this area to account for all of the commuters heading to this area, trying to avoid the higher levels of traffic.

There are also some pockets in the San Fernando Valley where poverty is also considered uneven. The San Fernando Valley is perceived as a middle-class and very suburban area. Although it is a very working-class area with industries such as logistics, manufacturing, and service, there are higher levels of poverty in some parts of Van Nuys, Panorama City, Pacoima, and North Hills. With jobs being spread out all over the San Fernando Valley, it is crucial to have public transportation options because people live in different parts of the Valley due to living costs, while well-paying jobs are in other parts of the Valley.

Finally, we can see that there is significantly less poverty in areas including Santa Monica, Pacific Palisades, Beverly Hills, Manhattan Beach, Hermosa, Redondo, and Malibu. These are areas that are considered very wealthy with excellent school districts with higher property values, lower density of people, and housing. These areas show that the physical geography often aligns with socioeconomic status and doesn't have as many lower-income residents living there or percentage of poverty. These wealthier individuals often own their own

forms of transportation, thus limiting the need for additional bus stops in these areas because there aren't as many people who need them for their commute.

Looking at the sheer concentration of Bus Stops, there aren't many where the areas of poverty are the highest. There is a pretty consistent concentration of bus stops in locations where the poverty index is between 0 and 200. This visualization demonstrates that communities with higher poverty levels have significantly less access to transportation infrastructure. With a R^2 value of 0.03, the scatter plot below shows that there is a weak correlation between poverty and the number of bus stops. This means that the people in these communities may not be as connected and may lose access to resources that most people take for granted. In order to make sure that there is equitable transportation, there needs to be targeted expansion based on where there is a higher density of the population within these higher poverty areas that do not have access. It is not necessary to add more in the areas that already have access, but we need to have them spread out across the community.



Calling for Action

The results of the analysis conclude that both Compton and East Los Angeles face significant transportation inequities that need immediate attention. These communities show very high concentrations of households without cars, low median household incomes, and elevated poverty levels. These factors indicate that the residents rely heavily on public transportation daily, yet they do not receive the level of bus stop density found in transit-rich areas such as Koreatown or Downtown Los Angeles. This imbalance limits mobility for the people who rely on transit the most.

To address these inequities, Los Angeles County should prioritize expanding and improving bus services in Compton and East Los Angeles. Strengthening transit in these areas would give residents better access to employment, education, and healthcare. The county should also focus on physical improvements, such as better lighting and safer crossings, to improve the usability and safety of transit stops. Targeted investment in Compton and East Los Angeles would improve transportation equity and better the overall quality of life for these households.

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